15 Examples of Rate Laws in Heterogeneous Catalysis

Toluene + H_2 --> Benzene + CH_4

$$-r_T = \frac{kP_{H_2}P_T}{1 + K_B P_B + K_T P_T}$$

NO + CO → products (used in diesel exhaust remediation)

$$-r_{NO} = \frac{kP_{NO}P_{CO}}{(1 + K_{NO}P_{NO} + K_{CO}P_{CO})^2}$$

Dehydration of butanol to butene: $CH_3CH_2CH_2CH_2OH$ --> $CH_3CH{=}CHCH_3$ + H_2O

$$-r_{Bu} = \frac{kP_{Bu}}{(1 + K_{Bu}P_{Bu})^2}$$ Bu = butanol

In these rate laws:

- *P* are partial pressures

  With units of atm, for example

  P = CRT

- *k* are rate constants

  With units of $\frac{\text{mol}}{\text{volume} \cdot \text{time} \cdot \text{pressure}^x}$ OR $\frac{\text{mol}}{\text{mass catalyst} \cdot \text{time} \cdot \text{pressure}^x}$

x = reaction order

The rate law has units of

$$\frac{\text{mol}}{\text{vol} \cdot \text{time}} \qquad \frac{\text{mol}}{\text{mass catalyst} \cdot \text{time}}$$

k depends on

temperature & activation energy

If you are given a value of *k* at one temperature and need to compute the value at a new temperature, you can use the

Arrhenius eq: $k(T) = k(T_0)\exp\left(\frac{\Delta E^{act}}{R}\left(\frac{1}{T_0} - \frac{1}{T}\right)\right)$

- *K* are

adsorption constant

With units of

inverse pressure

They indicate the extent to which species bind to the catalyst. $\uparrow K_j$ $\uparrow$ amount of species j on the surface

K depend on

temperature, heat of adsorption of species j

If you are given a value of *K* at one temperature and need to compute the value at a new temperature, you can use the

van't Hoff eq: $K(T) = K(T_0)\exp\left(\frac{\Delta H^{ads}}{R}\left(\frac{1}{T_0} - \frac{1}{T}\right)\right)$

Toluene + H_2 --> Benzene + CH_4

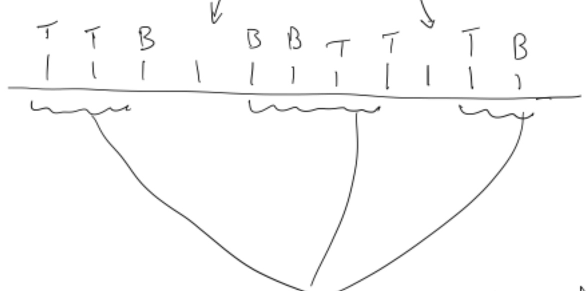
numerator: temperature & concentration dependence – analogous to elementary rate law

$$-r_T = \frac{kP_{H_2}P_T}{1 + K_BP_B + K_TP_T}$$

← indicates which species bind strongly and block sites on the catalyst

Benzene Toluene

"free" sites – reaction occurs here



BLOCKED – Cannot perform the reaction

NO + CO → products (used in diesel exhaust remediation)

$$-r_{NO} = \frac{kP_{NO}P_{CO}}{(1 + K_{NO}P_{NO} + K_{CO}P_{CO})^2}$$

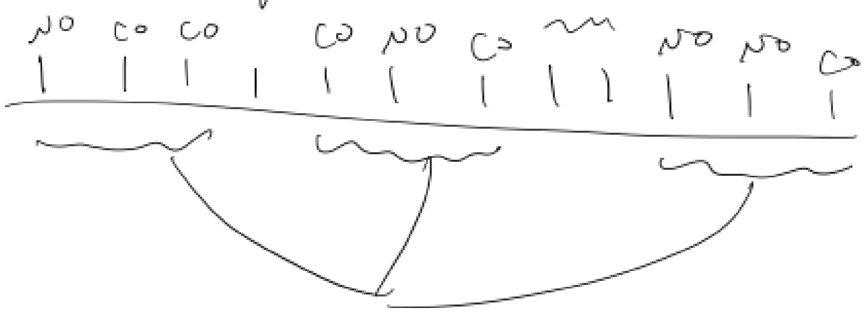
reaction requires two open sites

NO

CO

free but insufficient b/c the reaction requires 2 sites

reaction occurs here

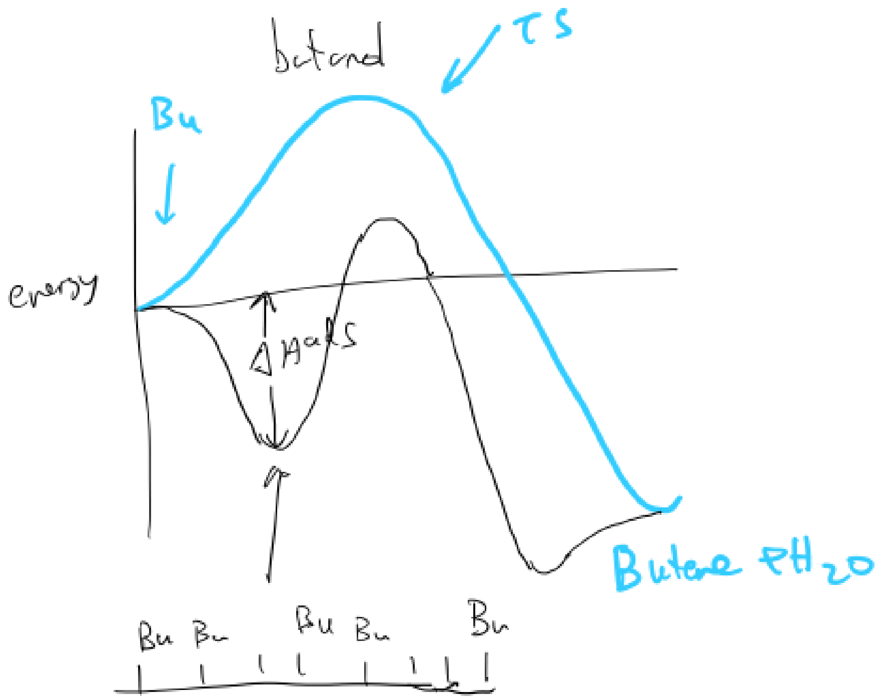


BLOCKED

Dehydration of butanol to butene: $CH_3CH_2CH_2CH_2OH \rightarrow CH_3CH{=}CHCH_3 + H_2O$

$$-r_{Bu} = \frac{kP_{Bu}}{(1 + K_{Bu}P_{Bu})^2}$$

butanol



the deeper the well the more Bu that binds / takes up sites on the catalyst